

Interactive / complex / 3

IC 1

IC 2

IC 3

IC 4

IC 5

IC 6

IC 7

IC 8

IC 9

IC 10

IC 11

IC 12

IC 13

IC 14v1

IC 14v2

query	Interactive / complex / 3				
title	Friends and friends of friends that have been to given countries				
pattern	The diagram illustrates the query pattern. It starts with a person: Person entity (orange box) with attribute <code>id = \$personId</code>. This person knows*1..2 otherPerson: Person entities (orange box). The otherPerson entity has attributes <code>id</code>, <code>firstName</code>, and <code>lastName</code>. The otherPerson entity hasCreator Message entities (purple box). There are two Message entities shown, each with a filter: <code>\$startDate ≤ creationDate < \$startDate + \$durationDays</code>. The Message entities isLocatedIn Country entities (yellow box). The Country entities are named <code>countryX: Country</code> and <code>countryY: Country</code>, with attributes <code>name = \$countryXName</code> and <code>name = \$countryYName</code> respectively. The Country entities isPartOf City entities (green box). The City entities have a filter: <code>«neg» isPartOf</code>.				
description	Given a start Person with ID <code>\$personId</code> , find Persons that are their friends and friends of friends (excluding the start Person) that have made Posts / Comments in both of the given Countries (named <code>\$countryXName</code> and <code>\$countryYName</code>), within <code>[\$startDate, \$startDate + \$durationDays)</code> (closed-open interval). Only Persons that are foreign to these Countries are considered, that is Persons whose location Country is neither named <code>\$countryXName</code> nor <code>\$countryYName</code> .				
params	1	<code>\$personId</code>	ID		
	2	<code>\$countryXName</code>	String	In SNB Interactive v2, this query has two variants: (a) Correlated Countries (b) Anti-correlated Countries	
	3	<code>\$countryYName</code>	String		
	4	<code>\$startDate</code>	Date	Beginning of requested period	
	5	<code>\$durationDays</code>	32-bit Integer	Duration of requested period, in days. The interval <code>[\$startDate, \$startDate + \$durationDays)</code> is closed-open	
result	1	<code>otherPerson.id</code>	ID	R	
	2	<code>otherPerson.firstName</code>	String	R	
	3	<code>otherPerson.lastName</code>	String	R	
	4	<code>xCount</code>	32-bit Integer	A	Number of Messages from Country named <code>\$countryXName</code> created by the Person within the given time
	5	<code>yCount</code>	32-bit Integer	A	Number of Messages from Country named <code>\$countryYName</code> created by the Person within the given time
	6	<code>count</code>	32-bit Integer	A	<code>count = xCount + yCount</code>
sort	1	<code>count</code>	↓		
	2	<code>otherPerson.id</code>	↑		
limit	20				
CPs	2.1, 3.1, 5.1, 8.2, 8.5				
relevance	This query looks for paths of length two and three, starting from a Person, going to friends or friends of friends, and then moving to Messages. This query tests the ability of the query optimizer to select the most efficient join ordering, which will depend on the cardinalities of the intermediate results. Many friends of friends can be duplicate, then it is expected to eliminate duplicates and those people prior to access the Post and Comments, as well as eliminate those friends from Countries named <code>\$countryXName</code> and <code>\$countryYName</code> , as the size of the intermediate results can be severely affected. A possible structural optimization could be to materialize the number of Posts and Comments created by a Person, and progressively filter those people that could not even fall in the top 20 even having all their posts in the Countries named <code>\$countryXName</code> and <code>\$countryYName</code> .				